Module - 4

DTFT

$z = e^{j\Omega}$.

$$x(n) \xrightarrow{DTFT} X(e^{j\Omega})$$

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\Omega n}$$

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega})\, e^{j\Omega n}\, d\Omega .$$

Problems:

① $x(n) = a^n u(n)$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} a^n u(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} (a e^{-j\omega})^n$$

$$\boxed{X(e^{j\omega}) = \frac{1}{1 - a e^{-j\omega}}} = \frac{1}{1 - a(\cos\omega - j\sin\omega)}$$

$$= \frac{1 e^{j0}}{(1 - a\cos\omega) + j(a\sin\omega)}$$

$$\boxed{\begin{array}{c} 2 e^{j40} \\ 3 + 2 e^{j40} \end{array}}$$

$$|X(e^{j\omega})| = \frac{1}{\sqrt{(1 - a\cos\omega)^2 + (a\sin\omega)^2}}$$

$$\angle X(e^{j\omega}) = 0 - \tan^{-1}\left(\frac{a\sin\omega}{1 - a\cos\omega}\right)$$

② $x(n) = \left(\frac{1}{2}\right)^n u(n-2)$

③ $x(n) = 2^{n+1} u(n+1)$

④ $x(n) = 3^n u(-n)$

⑤ $x(n) = (2)^n u(-n-1)$

⑥ $x(n) = a^{|n|}$

7, $x(n) = u(n) - u(n-8)$

8, $x(n) = 5\delta(n)$

9, $x(n) = 3\delta(n-2)$

10, $x(n) = \{2, \underset{\uparrow}{3}, 6\}$

Properties: $x(n) \longrightarrow X(e^{jn})$

Time shift

$x(n-n_0) \longrightarrow e^{-jn_0} X(e^{jn})$

freq shift

$e^{jn_0 n} x(n) \longrightarrow X(e^{j(n-n_0)})$

differentiation

$-jn\, x(n) \longrightarrow \frac{d}{dn} X(e^{jn})$

scaling

$x(an) \longrightarrow X(e^{jn/a})$

Convolution

$x_1(n) * x_2(n) \longrightarrow X_1(e^{jn}) \cdot X_2(e^{jn})$

Problems using Properties $DTFT\left[a^n u(n)\right] = \dfrac{1}{1-ae^{-jn}}$

1, $x(n) = \left(\frac{1}{3}\right)^{n-2} u(n-2)$

2, $x(n) = 2^n u(n-1)$

3, $x(n) = n\, a^n u(n)$ Ans: $\dfrac{ae^{-jn}}{(1-ae^{-jn})^2}$ $-jn\, x(n)$ Prop

4, $x(n) = e^{-j2n} (3)^n u(n)$

$$\left[3^n u(n)\right] = \frac{1}{1-3e^{-jn}}$$

$$\left[e^{-j2n} (3)^n u(n)\right] = \frac{1}{1-3e^{-j(n+2)}}$$

5, $x(n) = \cos \frac{\pi}{3} n\, a^n u(n)$

6, $x(n) = 2 \sin \frac{\pi}{4} n\, (3)^n u(n)$

7, $x(n)$ has DTFT $X(e^{jn}) = \dfrac{1}{1-ae^{-jn}}$

then find DTFT of

(a) $x(n+1)$ Ans: $\dfrac{e^{jn}}{1-ae^{-jn}}$

(b) $x(2n)$ $\dfrac{1}{1-ae^{-j(n/2)}}$

(c) $x(n) * x(n-1)$

(d) $e^{j\pi/2\, n} x(n+2)$

8, Find IDTFT of

$$X(e^{jn}) = \frac{1}{1 - \frac{1}{2} e^{-jn}}$$

Ans: $\left(\frac{1}{2}\right)^n u(n)$

9, $$X(e^{jn}) = \frac{e^{-j3n}}{1 - \frac{1}{3} e^{-jn}}$$

Ans: $\left(\frac{1}{3}\right)^{n-3} u(n-3)$

10, $$X(e^{jn}) = \frac{6}{e^{-j2n} - 5e^{-jn} + 6}$$

take $e^{-jn} = m$

11, $$X(e^{jn}) = \frac{3 - \frac{1}{4} e^{-jn}}{1 - \frac{1}{16} e^{-j2n}}$$

12, $$X(e^{jn}) = \frac{1}{(1 - a e^{-jn})^2}$$

using Convolution Property.

$$a^n u(n) * a^n u(n)$$

$$\sum_{k=-\infty}^{\infty} a^k u(k)\, a^{n-k} u(n-k)$$

$k \geq 0$ $\quad$ $n-k \geq 0$

$k \leq n$

$$\sum_{k=0}^{n} a^n = a^n \sum_{k=0}^{n} 1 = a^n (n+1)$$

Fourier Transform (CTFT / FT)

$$x(t) \longrightarrow X(j\omega)$$

FT $$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

IFT $$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega .$$

Problems:

1, $x(t) = e^{-at} u(t)$

$$X(j\omega) = \int_{-\infty}^{\infty} e^{-at} u(t)\, e^{-j\omega t}\, dt$$

$$= \int_{0}^{\infty} e^{-(a+j\omega)t}\, dt = \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_{0}^{\infty}$$

$$X(j\omega) = \frac{1}{a+j\omega}$$

$$|X(j\omega)| = \frac{1}{\sqrt{a^2+\omega^2}}$$

$$\angle X(j\omega) = -\tan^{-1} \frac{\omega}{a}$$

2, $x(t) = u(t)$ Ans: $\frac{1}{j\omega}$

3, $x(t) = u(t-2)$

4, $x(t) = t\, u(t)$

5, $x(t) = e^{-a|t|}$ Ans: $\frac{2a}{a^2+\omega^2}$

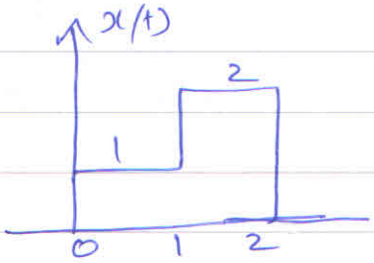
6, $x(t) = \begin{cases} A & -T/2 \le t \le T/2 \\ 0 & \text{otherwise} \end{cases}$ Ans: $\frac{2A \sin \omega T/2}{\omega}$

7, $x(t) = t\, e^{-at} u(t)$

8, $x(t) = \delta(t)$

9, $x(t) = 2\delta(t-1) + 3\delta(t) + 5\delta(t+2)$

10,



$$X(jw) = \int_0^1 1\, e^{-jwt}\, dt + \int_1^2 2\, e^{-jwt}\, dt$$

Ans. $X(jw) = \dfrac{1 + e^{-jw} - 2e^{-j2w}}{jw}$

Properties: $x(t) \longrightarrow X(j\omega)$

Time shift $x(t-t_0) \longrightarrow e^{-j\omega t_0} X(j\omega)$

Freq shift $e^{j\omega_0 t} x(t) \longrightarrow X(j(\omega-\omega_0))$

Time diff. $\frac{d}{dt} x(t) \longrightarrow j\omega X(j\omega)$

Freq diff. $-jt\, x(t) \longrightarrow \frac{d}{d\omega} X(j\omega)$

Scaling $x(at) \longrightarrow \frac{1}{a} X(j\omega/a)$

Convolution. $x_1(t) * x_2(t) \longrightarrow X_1(j\omega) \cdot X_2(j\omega)$

$$u(t) \longrightarrow \frac{1}{j\omega}$$

$$e^{-at}u(t) \longrightarrow \frac{1}{a+j\omega}$$

Problems using Property

1, $x(t) = 2e^{-3t}u(t)$ Ans $\frac{2}{3+j\omega}$

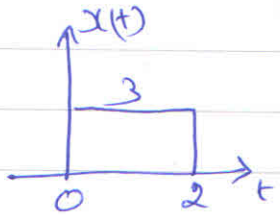
2, $x(t) = \frac{1}{2}e^{-2(t-3)}u(t-3)$

3, $x(t) = e^{-j\pi/2\, t}\, e^{-4t}u(t)$ Ans: $\frac{1}{4+j(\omega+\pi/2)}$

4, $x(t) = \cos \pi/3\, t\; e^{-at}u(t)$

5, $x(t) = t\,e^{-at}u(t)$ use $-jt\,x(t)$ Prop.

6,

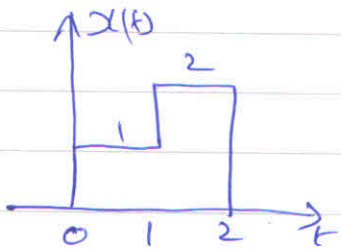


$$x(t) = 3\{u(t) - u(t-2)\}$$

$$\downarrow$$

$$\frac{3}{j\omega} - \frac{3e^{-j2\omega}}{j\omega}$$

7,



Find IFT

8, $X(j\omega) = \dfrac{e^{-j2\omega}}{a+j\omega}$ Ans $e^{-a(t-2)}u(t-2)$

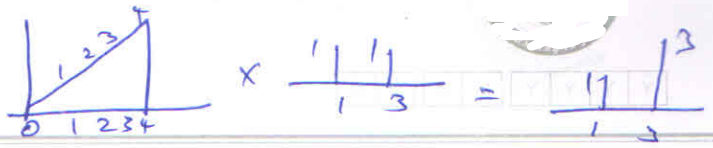
9, $X(j\omega) = \dfrac{\frac{1}{2} e^{j\frac{\pi}{3}\omega}}{\frac{1}{2}+j\omega}$

10, $X(j\omega) = \dfrac{2}{3+j(\omega-\pi/3)}$

11, $X(j\omega) = \dfrac{-j\omega}{(j\omega)^2+3(j\omega)+2}$ $m+1$ $A=1$

$m+2$ $B=-2$

12, $X(j\omega) = \dfrac{1}{(a+j\omega)^2}$ use convolution property.



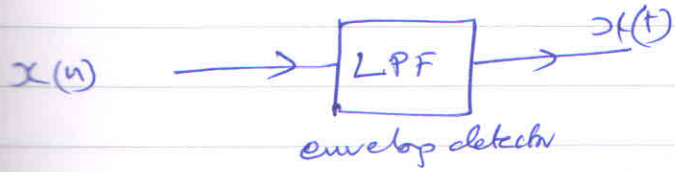
Sampling Theorem.

It is a technique to convert analog to digital signal (i.e) continuous to discrete signal. It is a process where we multiply analog signal $x(t)$ with train of impulse which result in digital signal $x(n)$

$x(t)$ $\quad \sum_{P=-\infty}^{\infty} \delta(t-P) \quad = \quad x(n)$

Over sampling

Under sampling

Right sampling $\qquad f_s = 2 f_m$

Reconstruction of signal